

KHADAMA, Russian Federation

WMO: 298920

Lat: 53.950N

Lon: 98.817E

Elev: 705

StdP: 93.14

Time Zone: 8.00 (E08)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.5	-33.7	-38.8	0.1	-36.4	-35.9	0.1	-33.4	5.5	-13.2	3.8	-12.2	0.9	170	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	27.5	16.5	25.1	15.7	23.0	15.0	18.2	24.2	17.1	22.7	16.1	21.1	1.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.6	20.0	15.2	11.7	18.8	14.2	11.0	17.6	54.5	24.0	51.1	23.0	47.8	21.0	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.0	3.0	2.4	-39.6	31.5	3.3	1.2	-42.0	32.4	-44.0	33.2	-45.8	33.9	-48.2	34.8		
(5)			-39.6	20.8	3.2	1.0	-41.9	21.5	-43.8	22.1	-45.5	22.7	-47.9	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-21.7	-16.7	-8.8	0.0	7.1	12.1	15.8	13.8	7.0	-0.2	-11.3	-20.3		
	DBStd	13.85	6.00	5.65	5.55	4.48	4.31	3.60	3.05	2.87	3.70	4.88	6.60	6.64		
	HDD10.0	4754	982	748	583	302	112	19	2	4	106	317	640	939		
	HDD18.3	7386	1240	981	842	552	349	188	88	142	341	575	890	1198		
	CDD10.0	421	0	0	0	0	22	82	181	122	14	0	0	0		
	CDD18.3	12	0	0	0	0	0	2	8	2	0	0	0	0		
	CDH23.3	403	0	0	0	0	46	97	197	57	6	0	0	0		
	CDH26.7	88	0	0	0	0	8	24	46	10	1	0	0	0		
(14) Wind	WSAvg	0.8	0.8	0.8	0.9	1.1	1.0	0.7	0.7	0.7	0.7	0.9	0.9	0.8		
(15) Precipitation	PrecAvg	525	5	4	9	23	53	89	119	115	64	19	11	7		
	PrecMax	675	14	8	17	41	85	135	197	188	137	29	28	15		
	PrecMin	442	1	1	1	9	29	44	66	51	36	6	4	1		
	PrecStd	60	3	2	4	8	19	22	28	39	25	7	6	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.8	1.3	11.8	18.3	27.9	29.5	30.2	27.8	24.0	17.0	6.5	-0.7		
		MCWB	-5.7	-3.0	4.7	8.6	14.3	16.5	18.2	17.5	14.3	8.7	1.3	-3.8		
	2%	DB	-7.2	-2.3	7.6	14.2	24.1	26.2	27.9	24.6	19.5	13.1	2.4	-5.1		
		MCWB	-8.5	-5.4	1.8	6.1	12.4	15.1	17.6	16.3	12.2	6.0	-0.9	-6.9		
	5%	DB	-10.0	-4.5	4.1	11.2	20.4	23.4	25.6	22.5	16.1	9.4	-0.4	-8.1		
		MCWB	-11.1	-6.8	-0.3	4.4	10.5	13.8	16.8	15.4	10.2	4.2	-2.7	-8.9		
	10%	DB	-12.6	-7.1	1.4	8.3	17.3	20.9	23.3	20.1	13.3	6.6	-2.4	-11.1		
		MCWB	-13.3	-8.9	-2.2	2.7	8.9	13.0	16.4	14.7	8.9	2.6	-4.0	-11.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.5	-2.6	4.6	8.8	14.7	18.4	20.1	18.5	15.2	8.9	1.9	-3.3		
		MCDB	-4.2	1.0	11.5	17.8	25.0	26.1	26.3	23.7	21.7	16.6	6.2	-1.2		
	2%	WB	-8.5	-5.2	1.8	6.2	13.0	16.2	18.8	17.2	12.7	6.5	-0.6	-6.9		
		MCDB	-7.2	-2.7	7.0	13.7	23.0	22.9	25.0	22.3	18.5	12.2	2.2	-5.6		
	5%	WB	-11.0	-6.8	-0.2	4.5	11.2	15.0	17.8	16.2	11.0	4.7	-2.5	-9.0		
		MCDB	-10.1	-4.6	3.7	10.7	18.8	21.0	23.6	21.1	15.2	8.5	-0.7	-8.0		
	10%	WB	-13.4	-8.8	-2.1	2.9	9.5	13.8	16.9	15.1	9.5	3.1	-4.0	-11.6		
		MCDB	-12.5	-7.1	1.3	7.6	16.3	19.3	22.2	19.3	12.7	5.9	-2.4	-11.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	17.3	18.6	13.9	14.5	14.5	13.4	12.1	12.4	12.8	10.9	10.0		
		MCDBR	13.3	19.7	21.6	19.7	22.4	20.8	18.8	17.3	18.4	18.0	12.1	11.5		
		MCWBR	11.8	16.5	15.8	12.0	11.6	10.6	9.3	9.1	10.7	11.1	9.0	10.0		
		MCDBR	12.9	19.5	20.6	18.3	19.8	17.0	15.9	15.2	17.5	16.6	11.6	11.5		
	5% WB	MCWBR	11.6	16.3	14.9	11.2	10.8	9.4	8.6	8.9	10.7	10.6	9.0	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.289	0.328	0.365	0.389	0.393	0.396	0.373	0.332	0.309	0.283	0.258		
	taud		2.162	2.148	2.146	2.230	2.245	2.343	2.370	2.408	2.468	2.405	2.216	2.110		
	Ebn at Noon		699	795	841	859	857	856	846	846	844	778	656	602		
	Edh at Noon		64	91	115	122	128	117	112	102	84	71	60	55		
(44) All-Sky Solar Radiation	RadAvg		1.02	1.97	3.38	4.64	5.11	5.54	5.03	4.37	3.16	1.91	1.02	0.70		
	RadStd		0.07	0.12	0.16	0.23	0.43	0.36	0.43	0.35	0.30	0.14	0.10	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page